

APOLLO MICROSYSTEMS · NSE

SCOUT ENGINE

Entry Scoring Architecture Reference

117

GRAND MAXIMUM SCORE

STOCK

APOLLO Hospitals (NSE)

CLASSIFICATION

5 Tiers

TIMEFRAMES

4H / Daily / Weekly

SCORING POOLS

3 Pools + 2 Bonuses

Scout Engine — Entry Scoring Architecture Reference

APOLLO Hospitals Enterprise Ltd. (NSE: APOLLOMICR) · July 2026

1. System Overview

The Scout Entry Engine evaluates APOLLO Hospitals stock across three timeframes (4H, Daily, Weekly) using a multi-pool scoring architecture. Entry signals are weighted, summed, and classified into actionable tiers. Table 1 summarizes the core parameters.

Parameter	Value
Stock	APOLLO Hospitals Enterprise Ltd. (NSE: APOLLOMICR)
Timeframes	4H, Daily, Weekly (no 3D)
Formula	Entry Score = $\min(100, \text{Pool A} + \text{Pool B} + \text{Pool C}) + \text{Bonus A} + \text{Bonus B}$
Grand Maximum	117 points
Backtest Period	January 2024 to June 2026
Output Files	scout_backtest_results.csv, scout_trade_events.csv

Table 1. System overview parameters.

2. Classification Thresholds

The total entry score maps to one of five classification tiers. Each tier prescribes a position sizing and stop-loss approach. Refer to Table 2 for the complete mapping.

Classification	Score Range	Action
No Entry	0 – 49	No trade
Pilot Buy	50 – 69	Small position, wide stop-loss
Standard	70 – 79	Medium position
Full Entry	80 – 89	Full position, tighter stop-loss
Full Strong	90+	Maximum conviction

Table 2. Score-to-classification mapping with prescribed actions.

3. Trough Detection

Troughs are the entry trigger mechanism. Each timeframe is scanned independently for local minima. Once a trough is identified, a 50-bar recovery window opens. Scoring is activated when **any** single timeframe is in its recovery window.

Detection condition (per TF): $RSI_{21} \leq 40$ and the current bar's low is the minimum over the preceding 11 bars (inclusive of current). This identifies a local price trough coinciding with an oversold RSI reading.

Post-trough recovery window: 50 bars after the trough bar. During this window, the TF is flagged as **in_recovery**. The scoring engine activates when **at least one** TF (4H, Daily, or Weekly) has this flag set to True.

4. Pool A — RSI Recovery (Maximum 44 Points)

Pool A measures how far RSI has recovered from oversold levels. It is divided into per-timeframe signals (scored on 4H, Daily, and Weekly independently) and cross-timeframe signals (scored once). Tables 3a and 3b detail each signal.

4.1 Per-Timeframe Signals

ID	Signal	Condition	Pts	Type	Example (Apr 8)
A1	RSI21 > 30	TF-RSI21 > 30	3	Dynamic	4H: 52.69=Y, D: 47.62=Y, W: 58.55=Y
A2	RSI21 > 40	TF-RSI21 > 40	3	Dynamic	4H: Y, D: Y, W: Y
A3	RSI21 > 45	TF-RSI21 > 45	2	Dynamic	4H: Y, D: Y, W: Y
A4	RSI36 > 30	TF-RSI36 > 30	2	Dynamic	4H: Y, D: Y, W: Y
A5	RSI56 > 30	TF-RSI56 > 30	2	Dynamic	4H: Y, D: Y, W: Y

Table 3a. Pool A per-timeframe RSI recovery signals. Max per TF: 12 pts. 3 TFs combined: 36 pts.

4.2 Cross-Timeframe Signals

ID	Signal	Condition	Pts	Example (Apr 8)
A6	Multi-Speed RSI Divergence	D-RSI21 > D-RSI36 > D-RSI56 (all three in strict ascending order)	4	D: 47.62 > 46.06 > 46.59 = NO (RSI56 > RSI36)
A7	RSI Recovery Momentum	D-RSI21 gained > 5 points over last 3 bars, and daily in_recovery=True	4	D: gain=47.62-43.97=3.65 in 3 bars = NO (< 5)

Table 3b. Pool A cross-timeframe signals (scored once, not per-TF).

*Pool A total = 36 (per-TF) + 8 (cross-TF) = **44 max points**.*

5. Pool B — Price Structure (Maximum 25 Points)

Pool B evaluates the relationship between price and key moving averages / targets. Most signals operate on the Daily timeframe. B1 and B2 are nested — if price is within 2% of TP28, both signals fire for a combined 8 points. See Table 4.

ID	Signal	Condition	Pts	TF / Type	Example (Apr 8)
B1	Price Approaching TP	D-Close within 5% below D-TP28	5	Daily / Dynamic	D: TP28=208.96, Close=208.05, gap=0.4% = Y

ID	Signal	Condition	Pts	TF / Type	Example (Apr 8)
B2	Price Near TP	D-Close within 2% below D-TP28	3	Daily / Dynamic	D: gap=0.4% = Y
B3	Price > WC50	D-Close > D-WC50	5	Daily / Dynamic	D: 208.05 < 222.82 = N
B4	WC50 Rising	D-WC50 > D-WC50 from 5 bars ago	4	Daily / Dynamic	D: WC50=222.82 > 222.27 (5 bars ago) = Y
B5	Price > MA200	D-Close > D-MA200	4	Daily / Dynamic	D: 208.05 < 243.64 = N
B6	Cross-TF Recovery Alignment	2+ TFs have RSI21 > 40 simultaneously	4	Multi / Dynamic	4H: 52.69>40=Y, D: 47.62>40=Y = Y

Table 4. Pool B price structure signals.

B1 and B2 are nested — if within 2%, both fire (5+3=8 pts).

6. Pool C — Confirmation (Maximum 15 Points)

Pool C provides confirmation through volume, price action, and indicator alignment. Most signals are simple binary checks on the Daily bar. C1 uses a fire-and-hold pattern (scored once per event). Refer to Table 5.

ID	Signal	Condition	Pts	TF / Type	Example (Apr 8)
C1	Multi-TF Trough Agreement	2+ TFs simultaneously in_recovery = 5 pts; 1 TF = 2 pts	2–5	Multi / Fire-hold	D=Y, 4H=Y, W=N → 2 TFs = Y (5 pts)
C2	Volume Above Average	D-Volume > 20-bar average D-Volume	3	Daily / Dynamic	D: 6.42M < 5.0M (20d avg) = N
C3	Bullish Close	D-Close > D-Open	2	Daily / Dynamic	D: 208.05 > 197.50 = Y
C4	RSI21 > RSI56	D-RSI21 > D-RSI56	3	Daily / Dynamic	D: 47.62 < 46.59 = N
C5	TP28 > WC50	D-TP28 > D-WC50	2	Daily / Dynamic	D: 208.96 < 222.82 = N

Table 5. Pool C confirmation signals.

7. Bonus A — Volume / Support / Trend (Maximum 15 Points)

Bonus A rewards high-conviction setups where volume, price support, and long-term trend align favorably. All three signals operate exclusively on Daily data. See Table 6.

ID	Signal	Condition	Pts	Example (Apr 8)
BA1	High Volume Spike	D-Volume > 2x 20-bar average	5	D: 6.42M vs 5.0M (20d avg) = N (< 2x)

ID	Signal	Condition	Pts	Example (Apr 8)
BA2	Price at Support	D-Close within 2% of 20-bar low	5	D: low_20=179.50, gap=(208.05-179.50)/179.50=15.9% = N
BA3	MA200 Slope Up	D-MA200 > D-MA200 from 20 bars ago	5	D: 243.64 > 243.44 (20 bars ago) = Y

Table 6. Bonus A volume, support, and trend signals.

8. Bonus B — Uptrend Correction (Maximum 18 Points)

Bonus B assesses whether the current dip is a correction within a broader uptrend — a high-reward scenario. All four signals use Daily data. Refer to Table 7.

ID	Signal	Condition	Pts	Example (Apr 8)
BB1	Above 50-bar Low	D-Close > 50-bar low	4	D: low_50=179.50, 208.05 > 179.50 = Y
BB2	Not Deeply Oversold	D-RSI21 > 35	4	D: 47.62 > 35 = Y
BB3	Higher Low Pattern	Recent 15-bar low > prior 15-bar low within 30 bars	5	Recent min=191.28, Prior min=179.50 → 191.28 > 179.50 = Y
BB4	Recovery Above MA200	Price was below MA200 within 20 bars, now above	5	D: was below at Mar 30 (182.16 < 243.44), now 208.05 < 243.64 = N

Table 7. Bonus B uptrend correction signals.

9. Score Composition Summary

Table 8 consolidates all five scoring components with their maximum point contributions, signal counts, evaluation timeframes, and architectural characteristics.

Pool	Max Pts	Signals	Evaluation TF	Architecture
Pool A (RSI Recovery)	44	7 (5 per-TF + 2 cross-TF)	4H, D, W (per-TF) + D (cross-TF)	Threshold levels, no momentum measurement
Pool B (Price Structure)	25	6	Primarily Daily	No 4H-level price signals
Pool C (Confirmation)	15	5	Primarily Daily, W for trough	Simple binary checks
Bonus A (Volume/Support)	15	3	Daily only	Volume and trend confirmation
Bonus B (Uptrend Correction)	18	4	Daily only	Contextual trend assessment
Grand Total	117	25	—	min(100, A+B+C) + Bonus A + Bonus B

Table 8. Score composition summary — all pools and bonuses.

10. Data Flow

The backtest script executes the following sequential steps to produce scored output:

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- 1.** Load 4H, Daily, and Weekly CSV files with pre-calculated indicators (RSI21, RSI36, RSI56, TP28, WC50, MA200, Volume).
 - 2.** Clean date columns and convert all numeric fields from string to float.
 - 3.** Detect troughs independently on each TF: $RSI_{21} \leq 40$ and 11-bar local minimum.
 - 4.** Set a 50-bar post-trough recovery window on each TF (in_recovery flag).
 - 5.** Filter Daily data to the backtest period (January 2024 onwards).
 - 6.** For each Daily bar, map to the corresponding 4H bar (last bar of the day) and the last completed Weekly bar.
 - 7.** Score the bar only if ANY TF has in_recovery = True; otherwise skip.
 - 8.** Pool A: Sum per-TF scores across 3 TFs, then add cross-TF signals (A6, A7).
 - 9.** Pool B, Pool C, Bonus A, Bonus B: Score primarily on Daily data.
 - 10.** Compute Base = $\min(100, A + B + C)$, then Total = Base + Bonus_A + Bonus_B.
 - 11.** Classify the total score into one of five tiers and write results to CSV plus a console summary.

This architecture reference documents every signal condition, its point value, the timeframe(s) evaluated, and verified example data from the April 8, 2026 bar on APOLLO Hospitals (NSE).